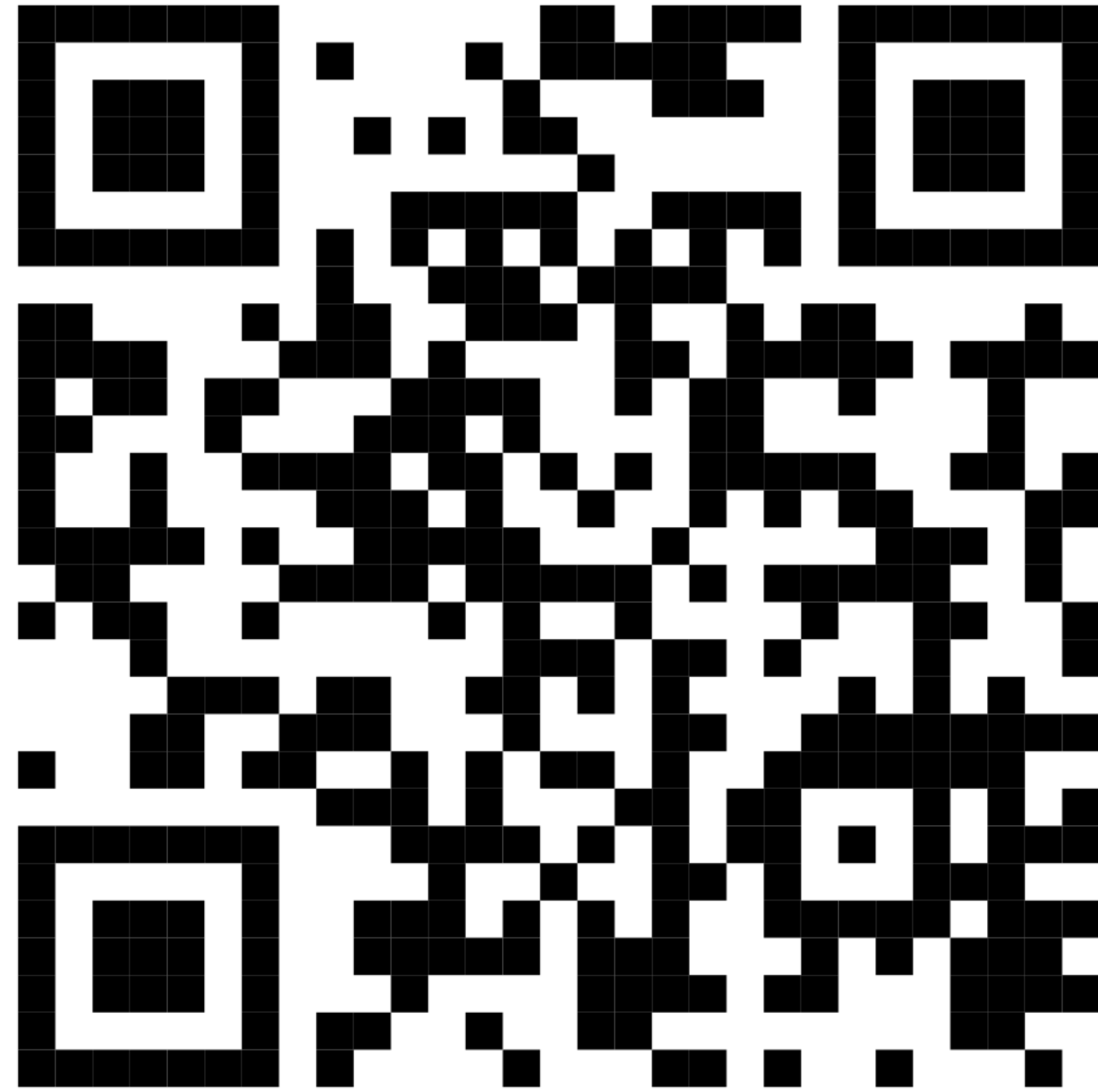




Robustness of random-walk Metropolis for steep potentials

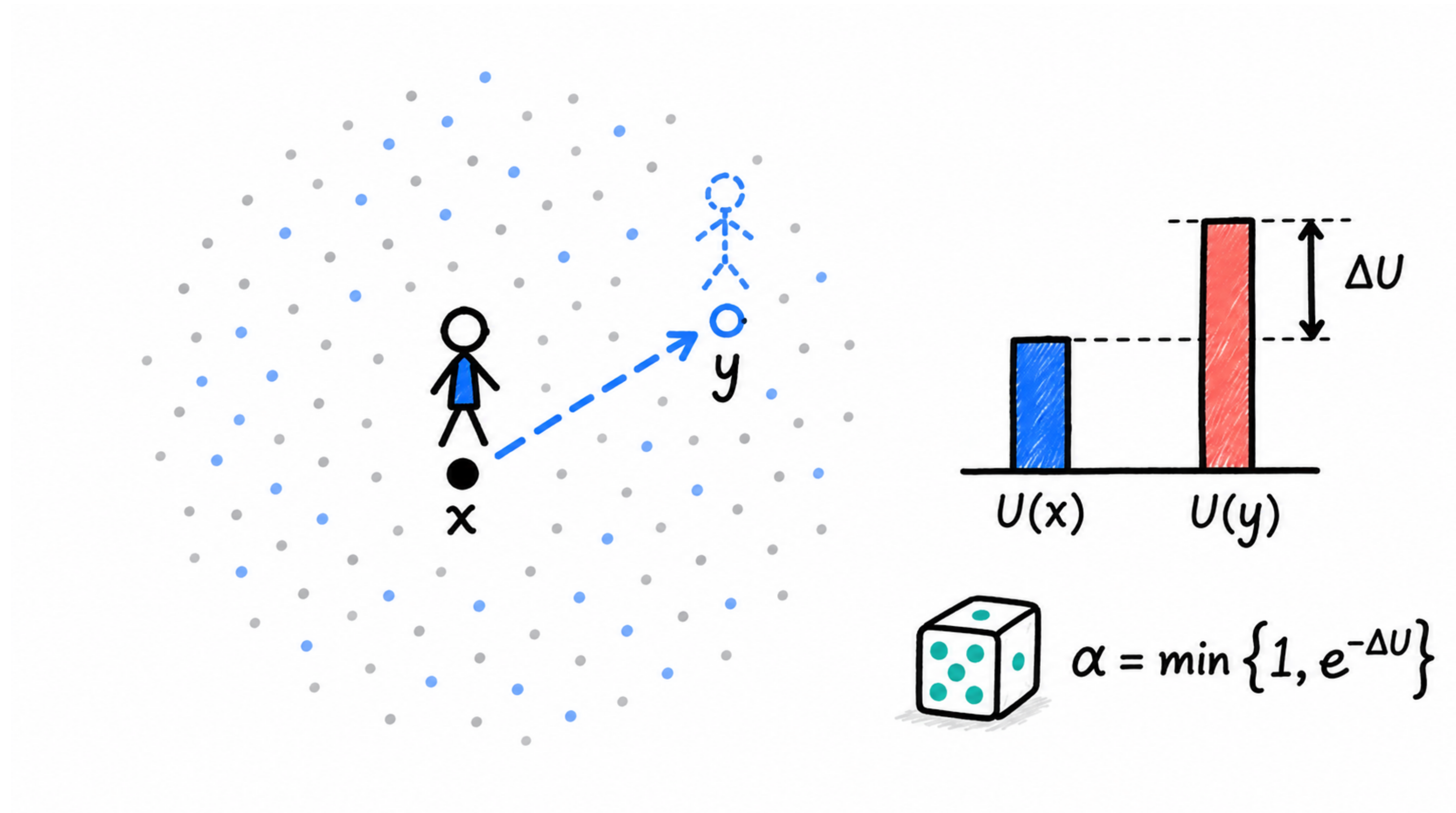
arxiv.org/abs/2608.20279



Steepness and Sampling

Theoretical Guarantees and Robustness

$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow X_N \stackrel{d}{\approx} \pi \propto \exp(-U)$$



Random Walk Metropolis

Guess-and-Check, with Guarantees

Random Walk Metropolis

- take $Q(x, dy) = \mathcal{N}(dy; x, \sigma^2 \cdot \mathbf{I}_d)$
- ‘accept’ moves (from Q) with probability

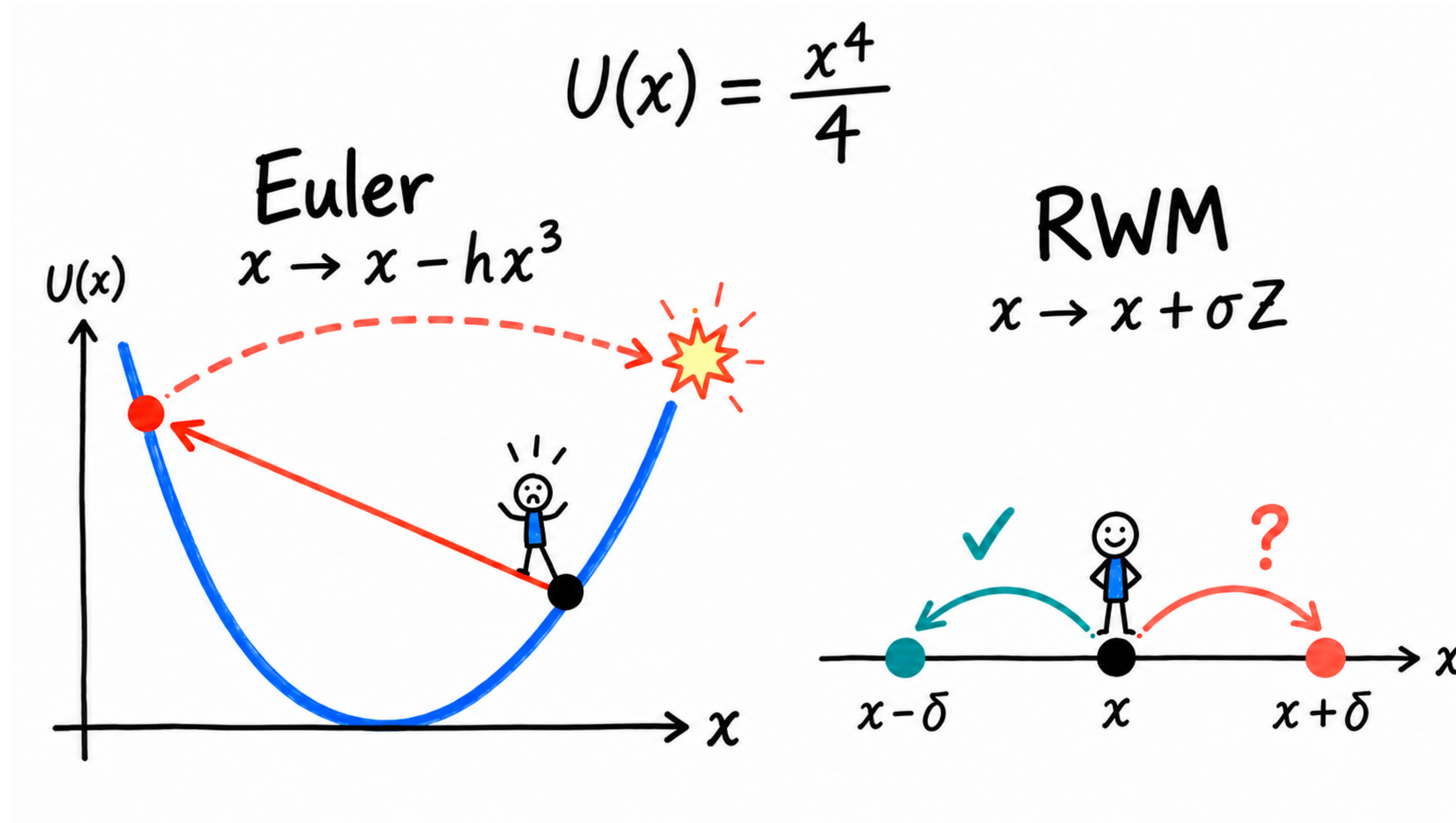
$$\alpha(x, y) = \min \left\{ 1, \frac{\pi(y)}{\pi(x)} \right\}$$

- this samples π correctly

RWM Guarantees

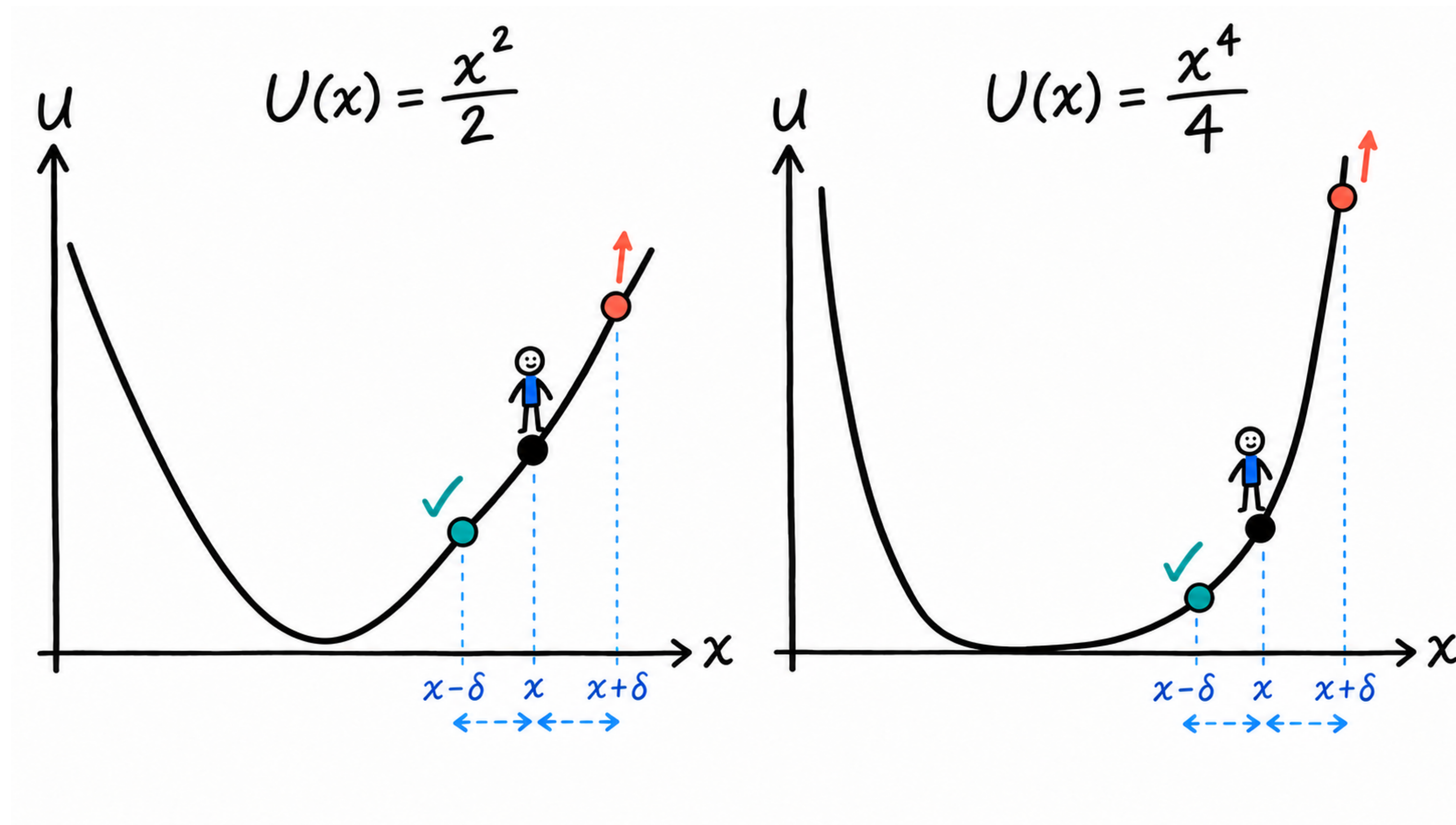
- understood very well; two key steps:
 - control $\inf_x \alpha(x) \geq \alpha_0 > 0$
 - study geometry ('isoperimetry') of target π
- for the first, we assume something like

$$\left\| \nabla U(x) - \nabla U(y) \right\| \leq \psi \left(\left\| x - y \right\| \right).$$



Gradient Instability

Sensitivity to Directional Information



Navigating Steep Landscapes

Superlinear Gradients and Failure of Taylor Expansions

Can we *establish* robustness?

Can we *explain* robustness?

Generalised Smoothness

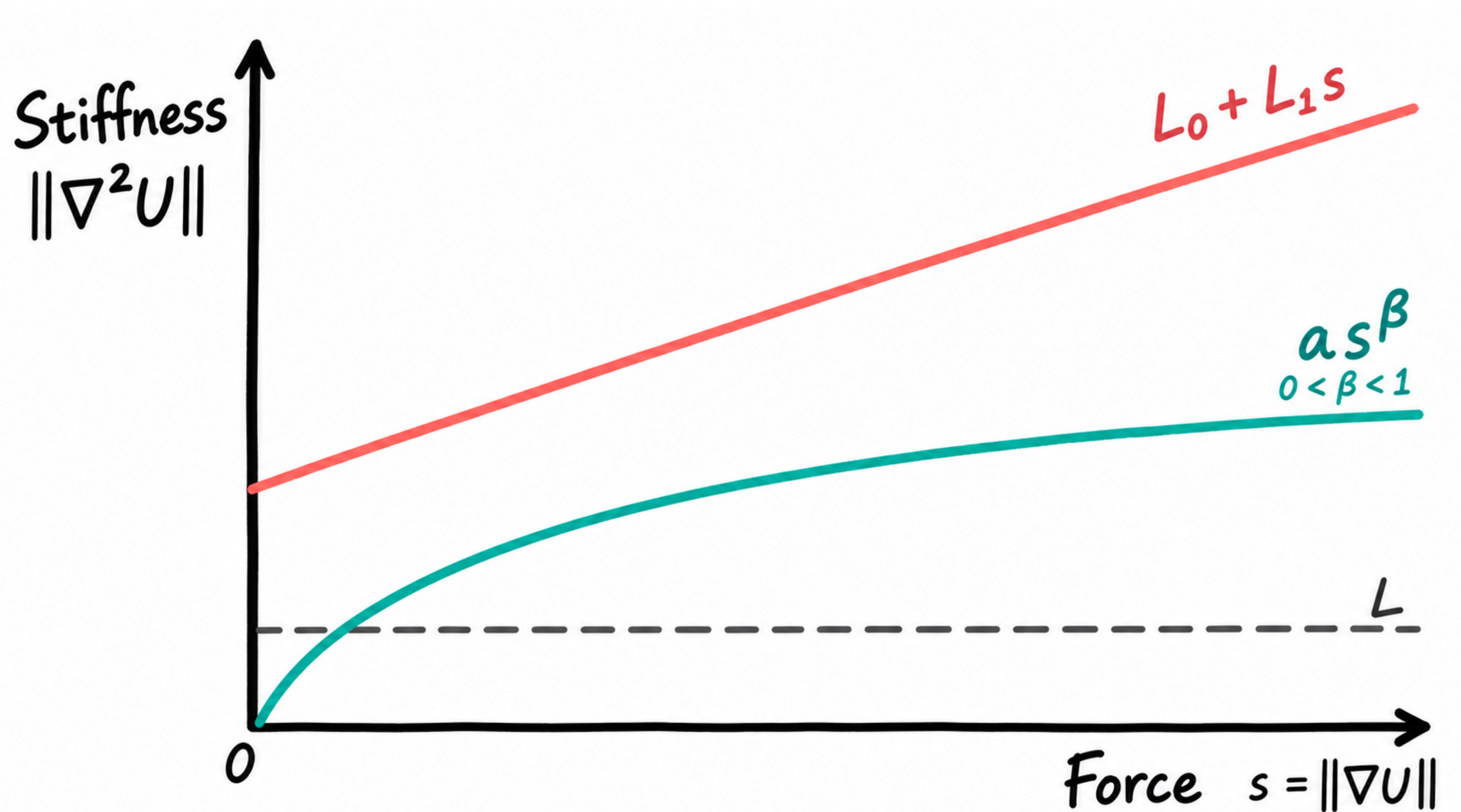
- what shall we do with unbounded stiffness:

$$\sup_x \left\| \nabla^2 U(x) \right\|_{\text{op}} = \infty \quad ?$$

- a solution: *self-bounding* stiffness:

$$\left\| \nabla^2 U(x) \right\|_{\text{op}} \leq G \left(\left\| \nabla U(x) \right\| \right)$$

with G sublinear

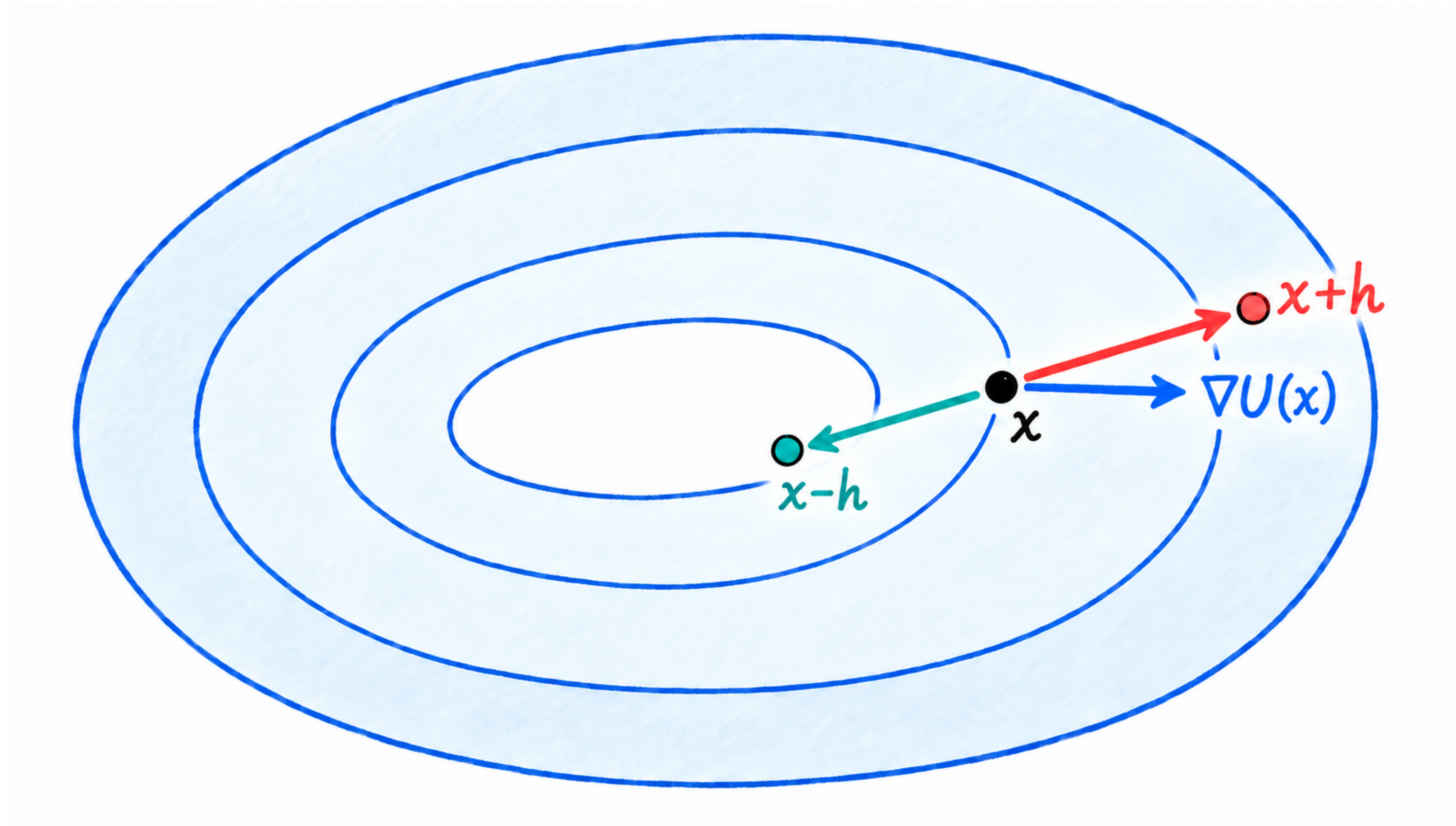


Controlling Stiffness by Force

Not Bounded, but 'Self-Bounding'

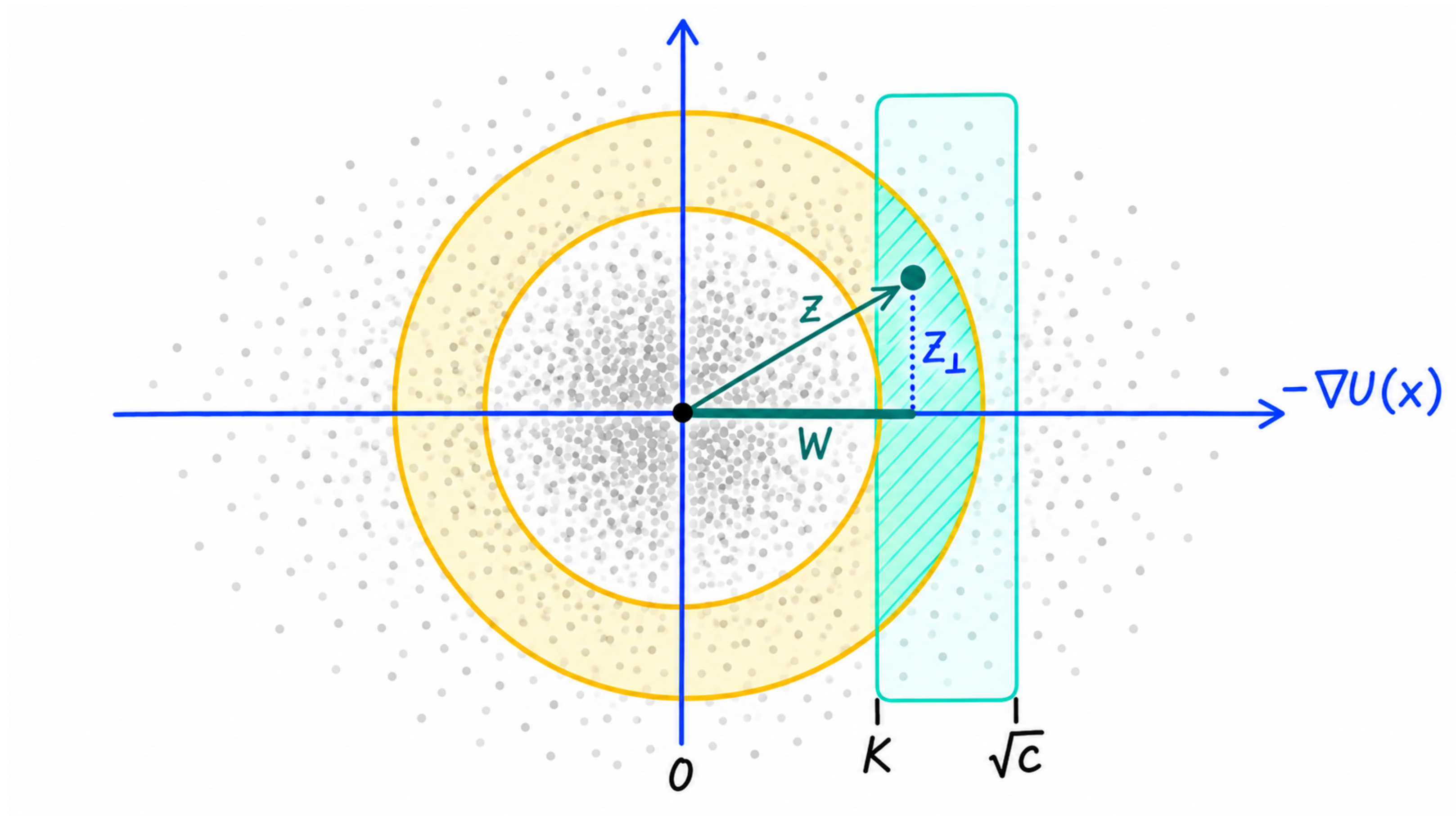
A Theorem

- Given a force-stiffness profile G ,
 - one can find explicit step-size $\sigma = \sigma_{d,G}$ such that for RWM,
 - $\inf_x \alpha(x) \geq \alpha_0 > 0$, and
 - $\sigma_{d,G}$ is only polynomially small with d
- $\rightsquigarrow$ *robustness!*



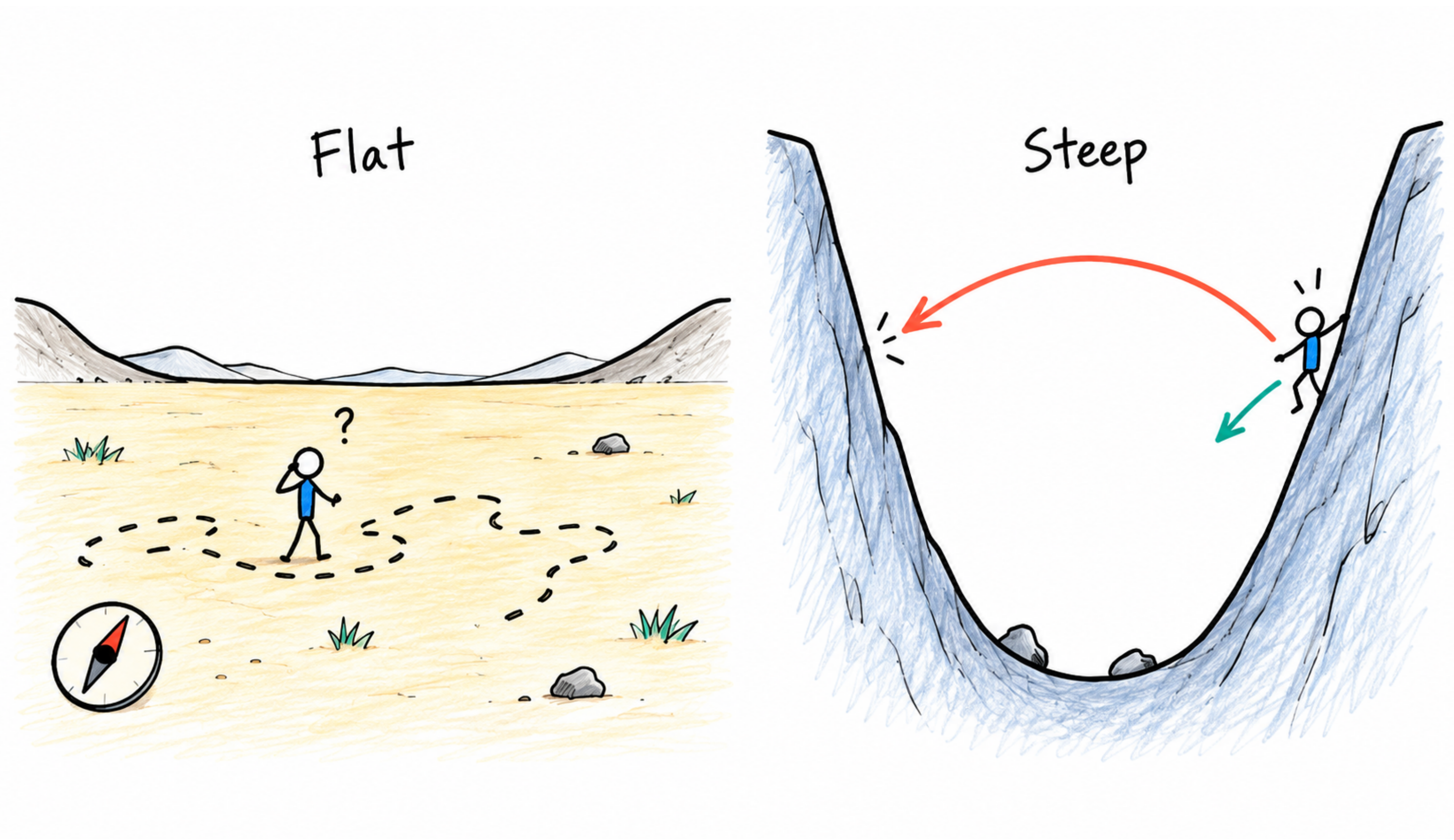
Hedging by Symmetry

One Favourable Direction Suffices



Moving in the Right Direction

Aim Downhill, Carefully



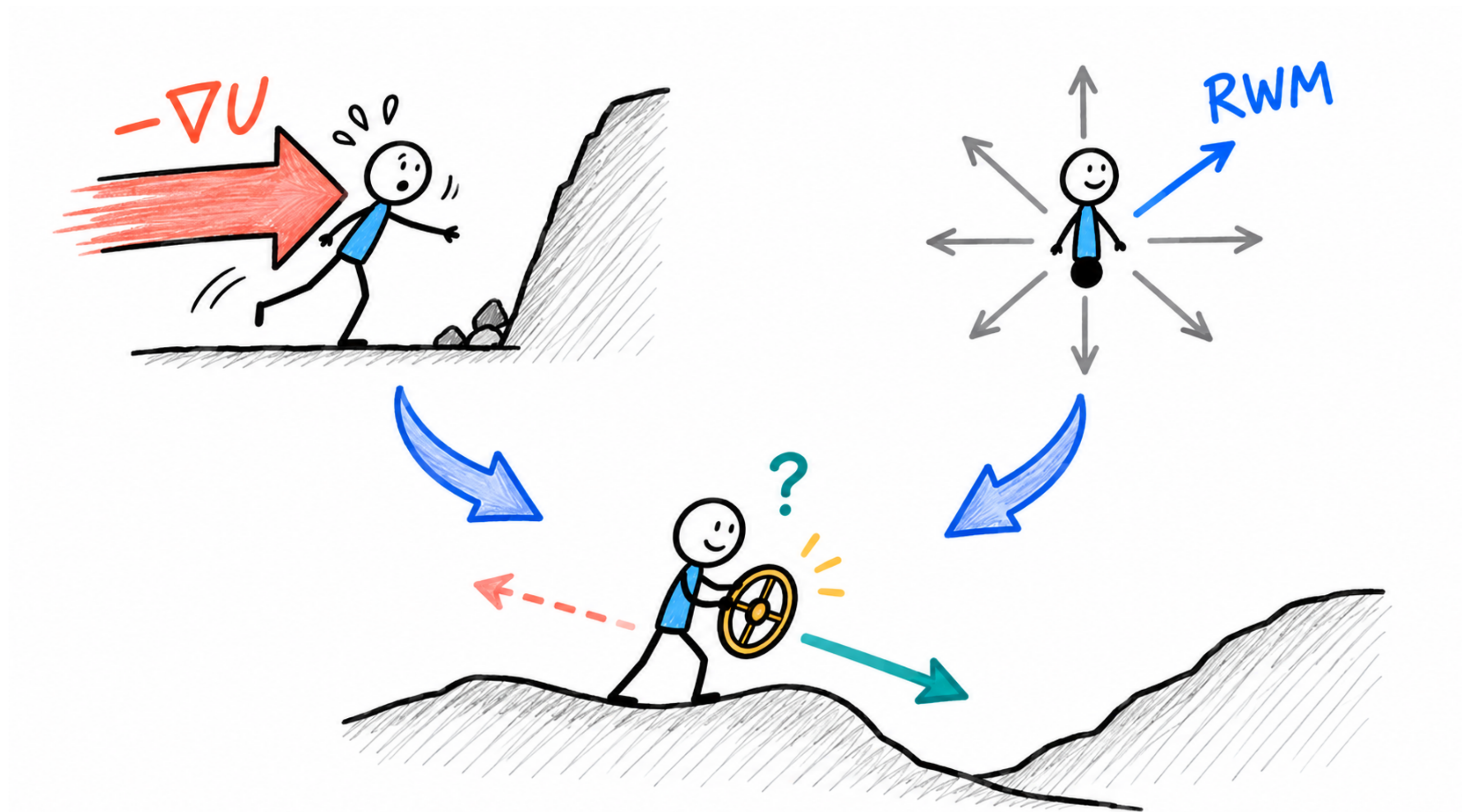
The Big Picture: *Robust Sampling*

Breakdown of 'Usual' MCMC Guarantees



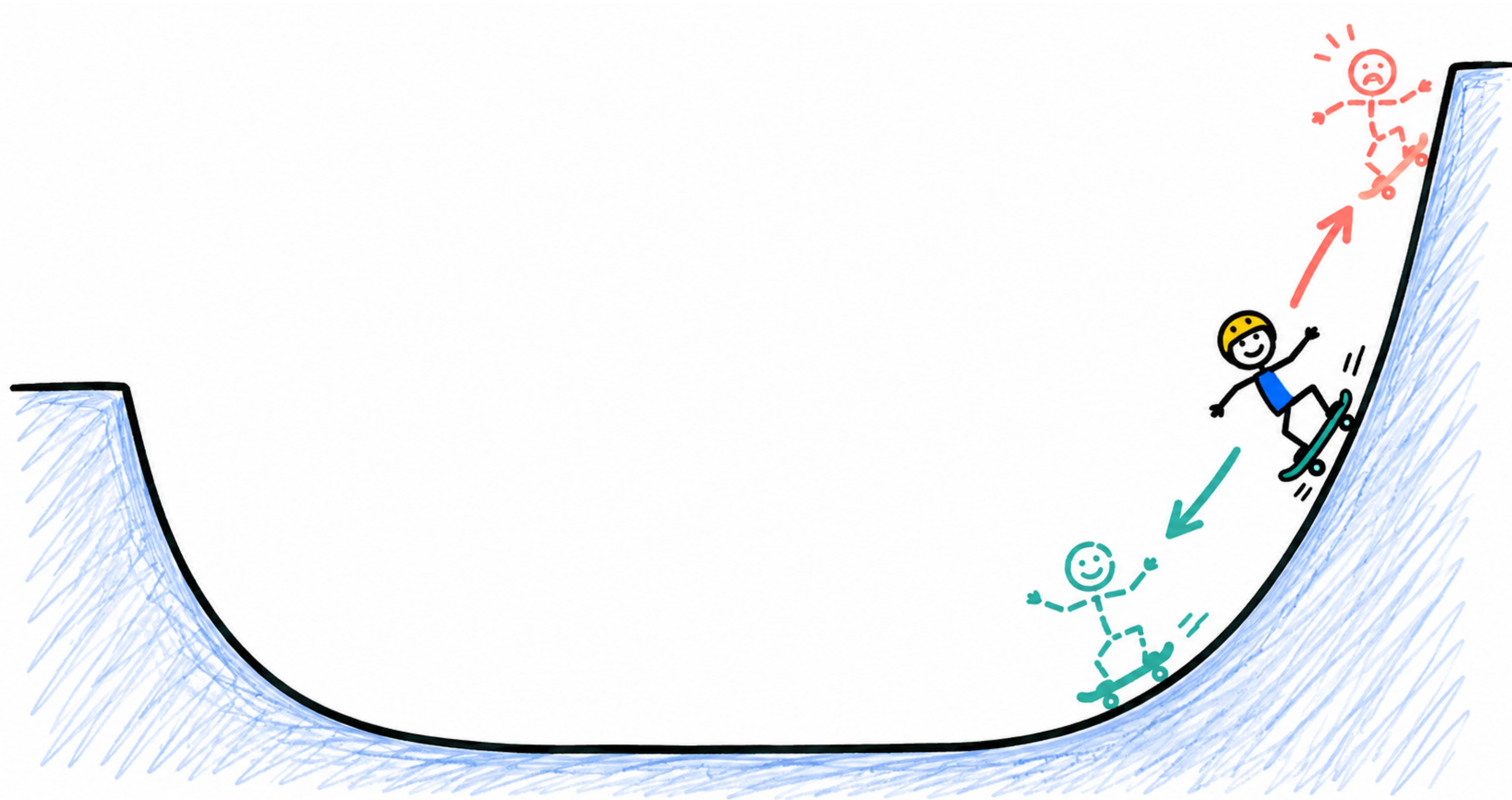
Some aspects of robustness in modern MCMC

<https://arxiv.org/abs/2511.21563>



How to Best *Steer* ‘Wild’ Gradients?

Sensitivity and Safety



Steepness *can* be Harmless

The Right Assumptions Clarify the Story